# Logic: switches that decide

## IoT · Level 2 · Lesson 2



# Today we will

- Describe AND, OR and NOT with switches and truth tables.
- Build AND and OR from two switches, in series and in parallel, with kit or on paper.
- Write a classroom rule (a fan, an alarm, a stair light) as logic.

# Three gates

- AND: all inputs on
- OR: any input on
- NOT: flips the input
- A truth table checks every case

# Human logic gates

1. Warm-up (5 min): When should the classroom fan run? Collect rules.
2. Teach (10 min): AND, OR and NOT with switches on the board.
3. Act it out (12 min): Pupils are the gates, with 1 and 0 cards.
4. Tables (8 min): Groups write the fan rule as a truth table.
5. Wrap-up (5 min): Share one real rule written as logic.

**Prepare 1 and 0 cards in advance; scrap paper is fine. If you have kit, wire the AND and OR circuits before class and test them.**

# Let’s discuss

1. Where do you see AND and OR rules at school or at home?
2. Why write a truth table before building anything?
3. Which classroom rule would you add NOT to, and why?

# Exit ticket

- Write the fan rule with AND, and fill in its four-row truth table.